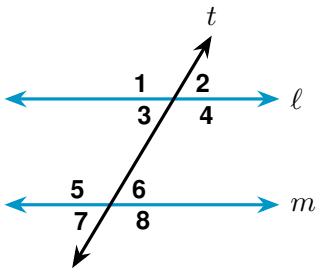
Parallel Lines & Transversals — Notes — Answer Key

Watch the video and fill in each blank. Pause when you see "Your turn."

Name: _____ Date: _____

☞ *Video: When the lines are parallel*

Key Idea



If $\ell \parallel m$:

Corresponding angles are **congruent**.

Alternate interior and alternate exterior angles are **congruent**.

Same-side interior angles are **supplementary** (add to $180°$).

☞ *Video: The easy way to see it*

Key Idea

Every angle is either **acute** or **obtuse**.

Two acute angles are **equal**. Two obtuse angles are **equal**. Acute + obtuse = $180°$.

☞ *Video: Find the angle*

Watch Me — Worked Example

$\ell \parallel m$, $m\angle 1 = 110°$. Find $m\angle 5$ and $m\angle 6$.

$\angle 1$ and $\angle 5$ are **corresponding** $m\angle 5 = 110°$

$\angle 5$ and $\angle 6$ are a **linear pair** $m\angle 6 = 180 - 110 = 70°$

☞ *Video: With algebra*

Watch Me — Worked Example

Alternate interior angles: $(3x + 15)°$ and $(5x - 25)°$.

$3x + 15 = 5x - 25$ alternate interior angles are **congruent**

$40 = 2x$, $x = 20$

angle $= 3(20) + 15 = 75°$ answer the question they asked

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☞ *Video: Watch out*

Common Mistake

Same-side interior angles are **supplementary**, not equal. Set them to add to $180°$.

☞ *Video: Your turn*

Your Turn

$\ell \parallel m$. Same-side interior angles are $(2x + 10)°$ and $(3x - 5)°$. Find x and both angles.

$(2x + 10) + (3x - 5) =$ 180 $\Rightarrow$ $5x + 5 = 180$ $\Rightarrow$ $x =$ 35

Angles: $80°$ and $100°$

Notice

Remember: Parallel lines: corresponding and alternate angles are **congruent**; same-side interior angles are **supplementary**. Acute = acute, obtuse = obtuse, acute + obtuse $= 180°$.

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